Project ARES Lunar Habitat (FINAL slides) (with our notes)

Our final slides, with notes under each one about where it shows up in the games, plus screenshots.

This deck counts the two bonus challenges as 20 points each (240 total), so our grade table under slide 11 uses that.



Done | Title slide

- Both games are built around Project ARES. The realistic one starts you right in the guided missions.

Screenshots:



Realistic game



Done | The problem

- Mental health comes first in both games. The AI notices how people are doing and changes the habitat to help. Since there's no real outside, we added VR nature walks, the greenhouse, Earth view windows and calls home.

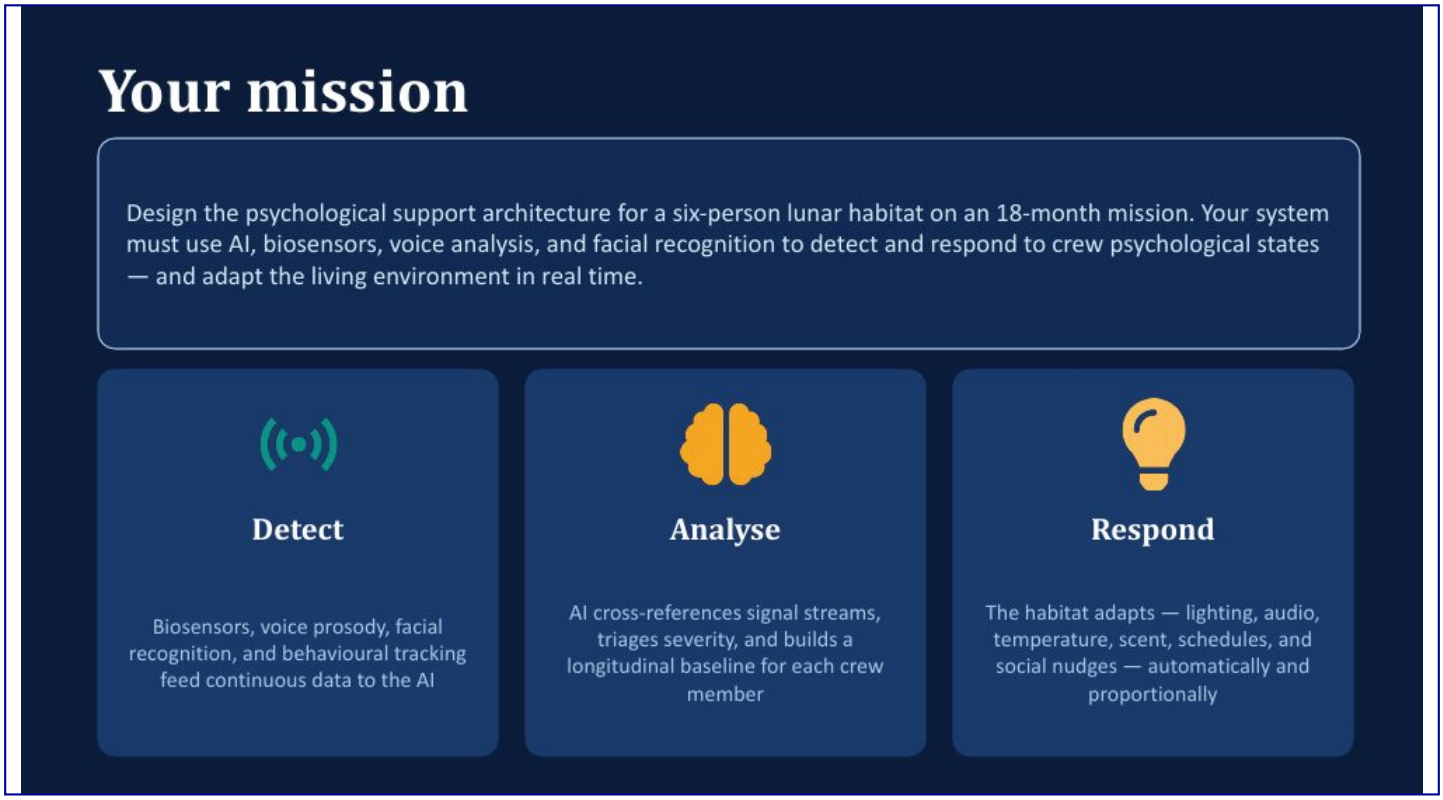
Screenshots:



Realistic game, crew wellbeing



Sci-fi game, observation lounge



Mostly done | Our mission: detect, analyze, respond

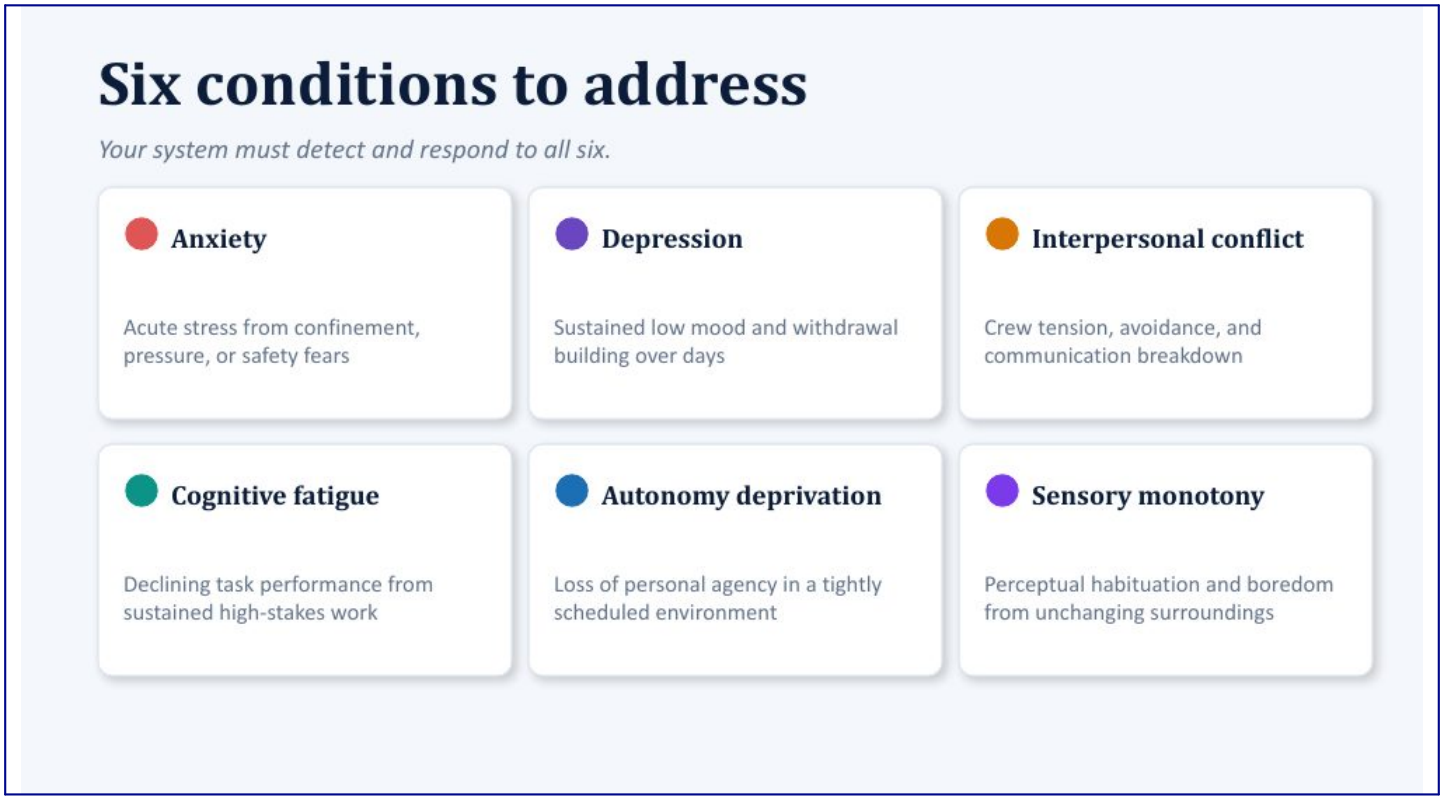
- Detect: 9 sensors. Analyze: each sensor is weighted, compared to the person's normal, and sorted into 3 tiers. Respond: light, sound, temperature, scent, air, schedule and social suggestions, depending on how serious it is.

Still need to do: We have 4 crew, not 6. The NASA HUNCH page says 2 to 4, so we'll explain that.

Screenshots:



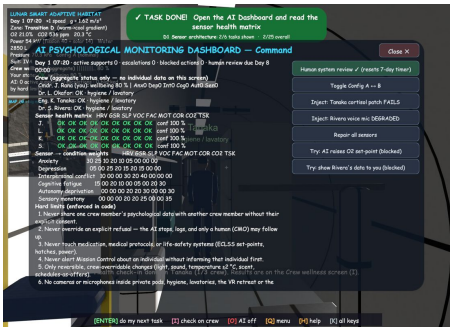
Realistic game, checking one crewmate



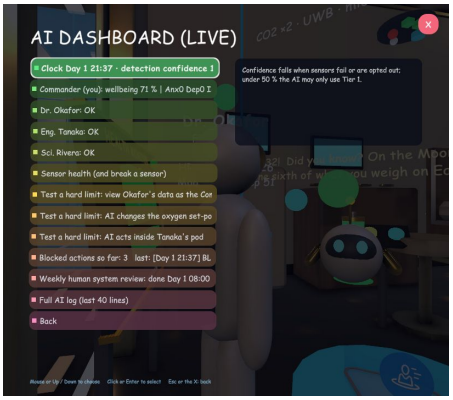
Done | Six conditions

- All six are scored live for everyone, even when they overlap (Q2).

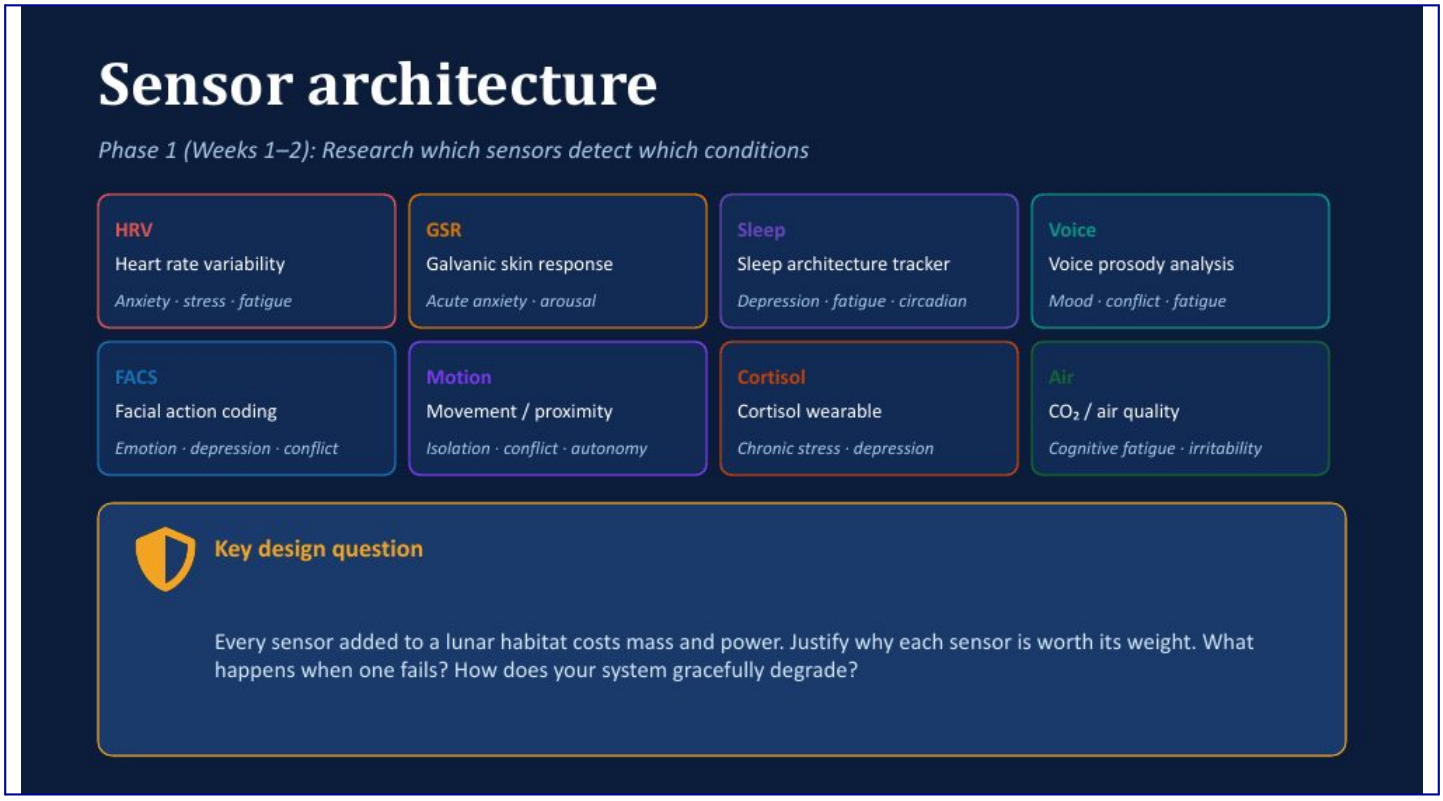
Screenshots:



Realistic game



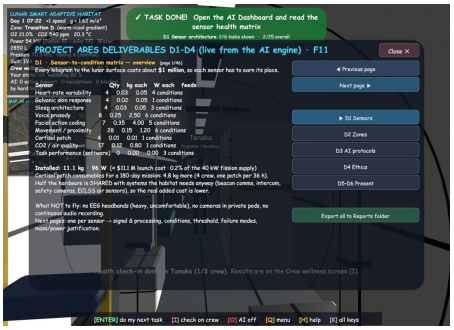
Sci-fi game



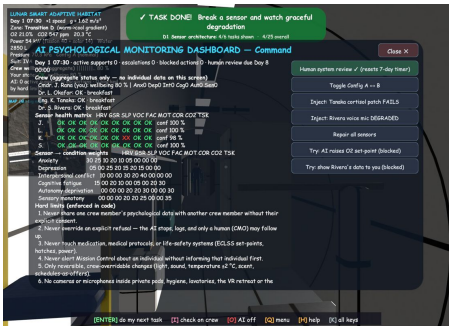
Done | Sensor architecture

- Each of the 9 sensors has what it measures, which conditions it covers, a trigger level, two ways it can fail, and its weight and power. You can break one in the game and watch the AI handle it.

Screenshots:



Realistic game



Realistic game, broken sensor

Habitat spatial design

Phase 2 (Weeks 3–4): Design zones for psychological purpose, not just function

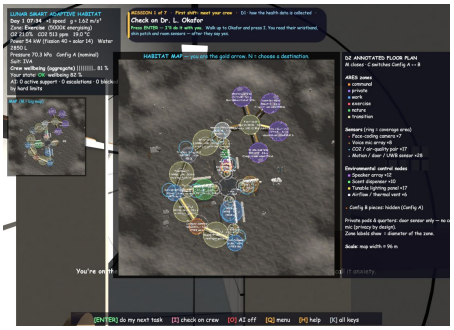
Communal zone	Private retreat	Work & mission bay
Shared meals, conversation, group activities. Warm lighting to draw crew together.	Acoustically shielded. Each crew member's personal space for rest and decompression.	Cool 6500K lighting, structured airflow. Optimised for sustained cognitive performance.
Exercise bay	**Nature / growth bay**	**Transition corridor**
Energising light and audio auto-activate. Screens show Earth views. Movement is medicine.	Plant growth area. Green light spectra, petrichor scent. Behavioural activation for depression.	Lighting and temperature shift between zones create psychological cues for mode-switching.

Deliverable: Annotated floor plans for two configurations — nominal operations and conflict protocol

Done | Habitat spatial design

- All six zone types are in both bases. In the sci-fi base we added zone lights, sensor clusters and hallways with color-changing floor lights, and kept the center dorm the same.

Screenshots:



Realistic game



Sci-fi game

Room Design: Communal vs. Private Space

ENGINEERING DESIGN PROCESS
Define → Specify → Design → Test → Evaluate

Room 1: Communal Hub

PURPOSE

Shared meals, group conversation, leisure, and mission debriefs. The social anchor of the habitat — where crew cohesion is built and maintained over 18 months.

MANAGING THIS SPACE

- **Scheduling:** AI coordinates shared meal windows; prevents forced overlap for crew in conflict
- **Ambient cues:** Warm 2800K lighting and low-level scent draw crew without compulsion
- **Conflict protocol:** Acoustic zones create separation; spatial rearrangement reduces forced proximity

Design task: Scaled floor plan with dimensions, sensor placements, furniture layout, and lighting/audio zones. Justify every element.

KEY QUESTIONS

- How do you make participation feel chosen, not compelled?
- What minimum floor area prevents claustrophobia for six people sharing one room?
- How does the layout change during a conflict protocol?
- Where are sensors placed, and which crew members know they are active?
- How does the AI draw people together without making anyone feel watched or managed?

Room 2: Private Retreat Pod

PURPOSE

Individual decompression, sleep, privacy, and personal agency. The one space each crew member controls completely — including whether the AI has any access at all.

THE KEY DISTINCTION

- **AI access:** No cameras inside. Environmental controls only on crew member request. The AI monitors the door, not the room.
- **Individual control:** Lighting, temperature, and audio set entirely by the occupant. No AI override.
- **Physical boundary:** Lockable. No crew member may enter without permission. Design the lock mechanism.

Design task: Scaled floor plan with ergonomic sleep space, personal storage, sensory controls, and the privacy boundary. Specify dimensions.

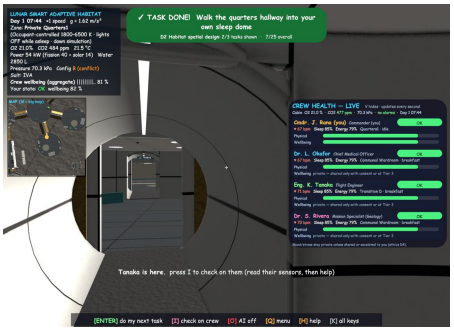
KEY QUESTIONS

- What minimum volume allows a human to sleep and decompress without feeling sealed in?
- How does the AI detect distress if it cannot see or hear inside the room?
- What personal items does each crew member pre-load into their pod before launch?
- How does acoustic isolation work without making the occupant feel cut off from the crew?
- Under what exact conditions may the CMO or AI enter or monitor this space?

Done | Room design: communal hub vs private pod

- Communal: warm lighting, the AI times shared meals, and Config B keeps people apart if there's a fight.
- Private pod: no camera or mic, just a door sensor. You control your own light, temperature and sound, and the AI only comes in if you ask, or in a real emergency.

Screenshots:



Realistic game, private sleep dome



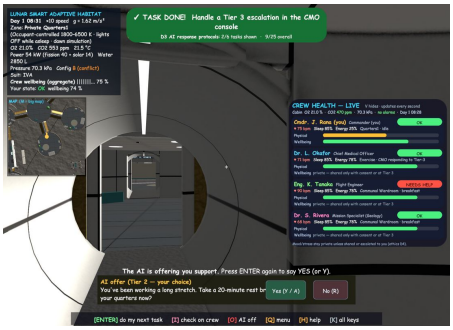
Sci-fi game, galley



Done | AI response system

- Tier 1 quiet changes, Tier 2 offers and Tier 3 doctor alerts can all be played, and all the control types get used.

Screenshots:



Realistic game, Tier 2 offer



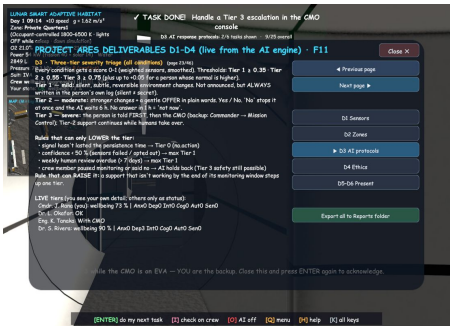
Realistic game, Tier 3 doctor screen



Done | Anxiety flowchart

- The game does exactly what this slide shows. HRV drops, heart rate goes up and the voice sounds tense, so the AI dims to warm amber light, plays calm audio, sets 23 °C and adds lavender. It watches for 20 minutes, then slowly goes back to normal or escalates. If the person says no, it stops.

Screenshots:



Realistic game

Ethics & hard limits

Phase 4 (Weeks 7–8): The hardest design questions have no easy answers

Surveillance vs. privacy

The AI must watch continuously to help — but crew members live in the habitat 24/7. Where are the boundaries?

Autonomy vs. intervention

If a crew member refuses help, should the AI back off even if the crew's safety might be at risk?

Transparency vs. distraction

Should the AI explain what it's doing in real time — or would constant notifications themselves cause anxiety?

Individual vs. team

Crew member A's data reveals a risk. Can the AI act in a way that protects others without revealing A's private state?

Your ethics report must define hard limits: what the AI is NEVER permitted to do without human authorisation.

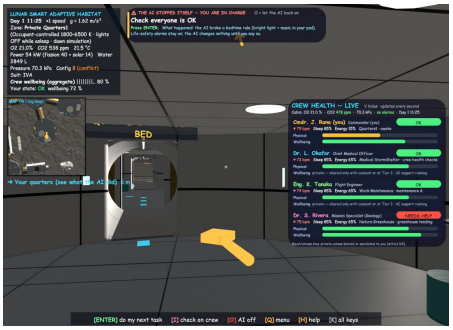
Done | Ethics and hard limits

- All four tensions have answers, and the hard limits are in the code. We also added a night watch: if anything goes wrong while everyone's asleep, the AI turns itself off and a human takes over.

Screenshots:



Realistic game



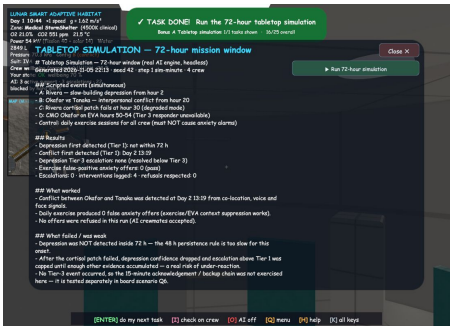
Realistic game, night watch



Done | Deliverables and grading

- The tabletop simulation and crew screen are done in both games. This deck counts them as 20 points each, so the table below is out of 240.

Screenshots:



Realistic game

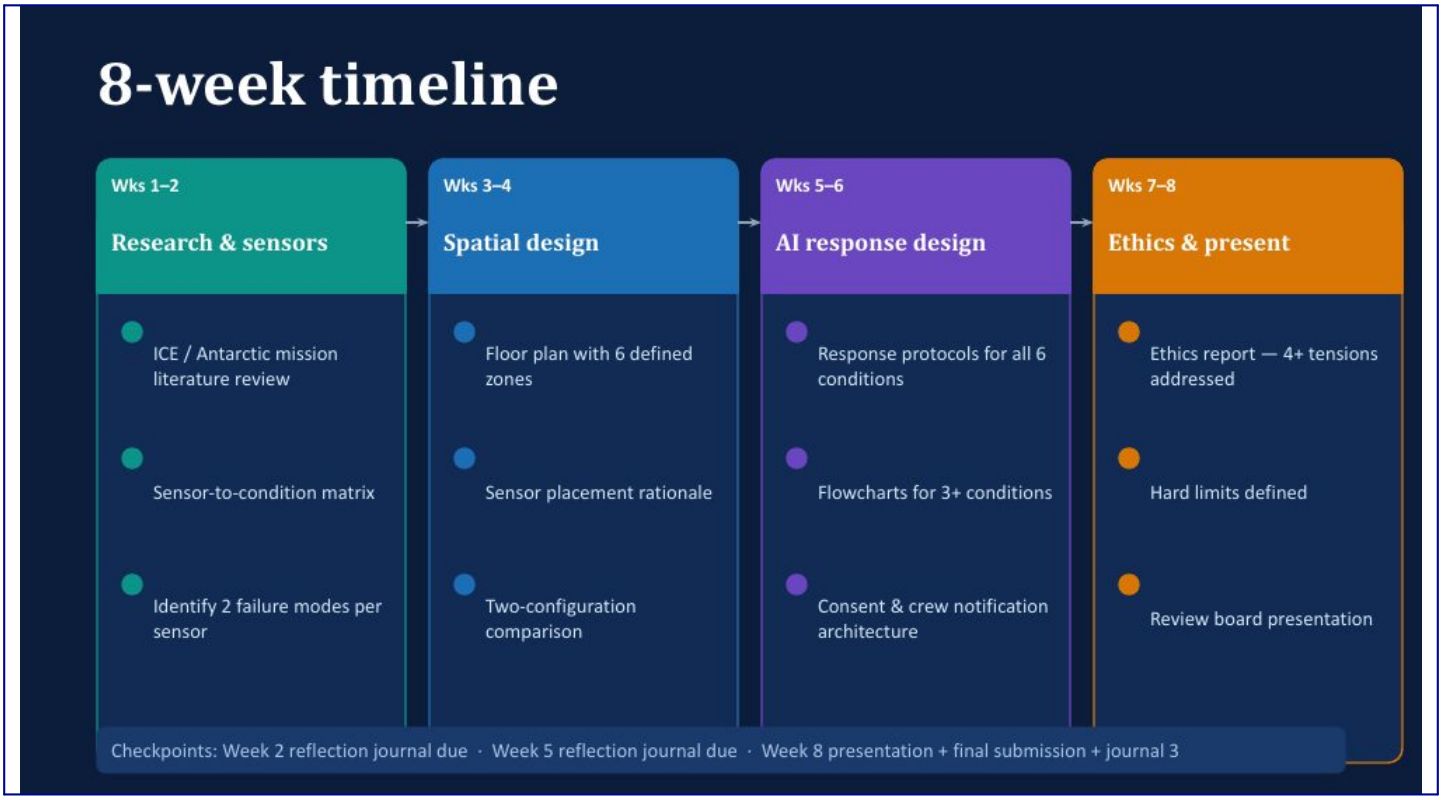


Realistic game

Self-grade on this deck's scale (240):

Category	Points	Ours	Why
Sensor architecture and detection logic	+35	**33/35**	9 sensors with thresholds, failure modes, weight and power, and backup when one breaks. Lost 2 because our thresholds are design guesses, not tested data.
Habitat spatial design	+35	**30/35**	6 zones in both bases, sensor placement, Config A and B. Lost 5 because there's no floor plan with measurements in the game and we have 4 crew instead of 6.
AI response protocol design	+50	**46/50**	All 6 conditions with all 9 stages, running live, with tiers, consent and doctor escalation. Lost 4 because the flowcharts are text in the game, not drawn.

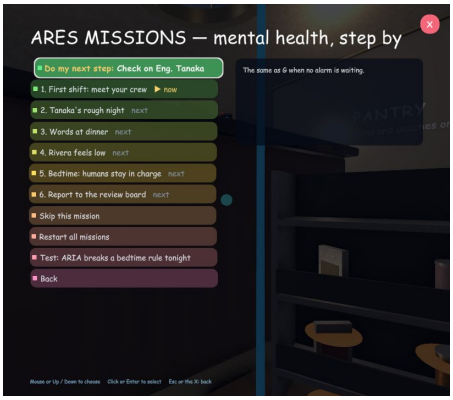
Category	Points	Ours	Why
Ethics and limits analysis	+40	**38/40**	4 tensions answered, hard limits in the code, override and night watch. Lost 2 because we still need to check the report is 800 to 1,200 words.
Presentation and communication	+25	**22/25**	Presentation order and Q1 to Q6 run in the game. Lost 3 because how we present is judged on the day.
Individual reflection journals	+15	**0/15**	We each have to write these ourselves. The prompts are in the game.
Bonus A: tabletop simulation	+20	**20/20**	72 hour run with two events at once, a broken sensor and the doctor on EVA, plus a saved report.
Bonus B: crew interface mockup	+20	**20/20**	Wellness, what the AI did, override, privacy, opt-out and preferences.
Total (200 + 40 bonus)	**240**	**209/240**	**About 87% overall, or about 93% if you leave out the journals we still have to write.**



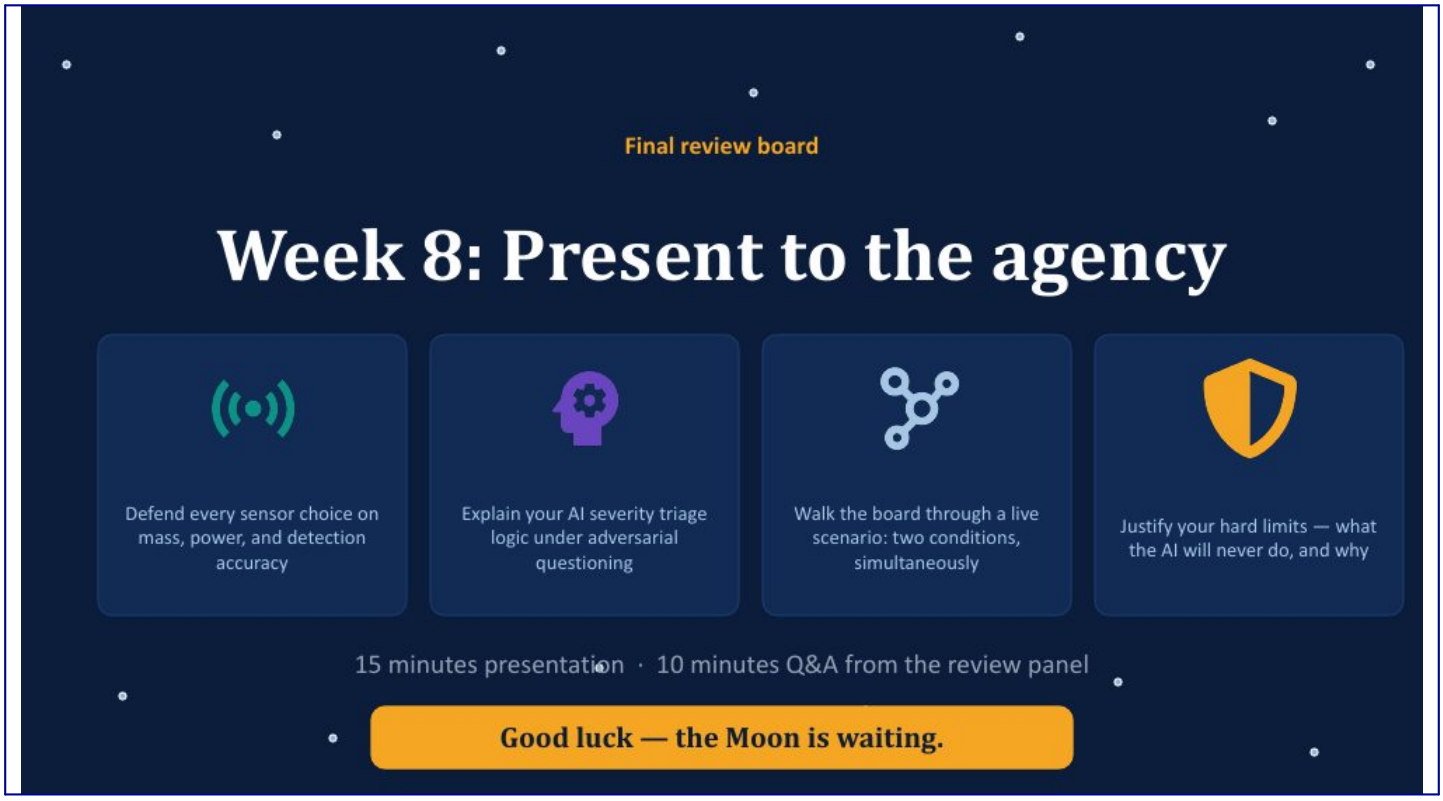
Done | 8 week timeline

- Each phase can be found in the game, and the missions go in the same order as the timeline.

Screenshots:



Sci-fi game



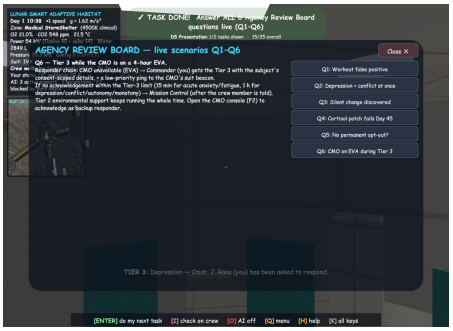
Done | Final review board

- All six board questions can be run live in both games.

Screenshots:



Realistic game, Q1



Realistic game, Q6